

Cybersecurity Concepts Lab for Teachers

Trust your hackles. Pause · Check · Tell.

Trust your hackles. Pause · Check · Tell.

FIELD COPY

1



Lock It

PASSWORDS AND ACCOUNTS

2



Spot It

SCAMS AND FAKES

3



Share Smart

PRIVACY

4



Safe People

WHO TO TRUST

5



Kind and Well

KINDNESS AND BALANCE

6



Fix It

TELL, REPAIR, RECOVER

PLATE I. The six strands, hand-inked, in the order the lessons take them.

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OCTOBER 2026

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Demo 13: Incident Response Drill and the Hotwash (Detect, Contain, Recover, Learn)

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Status: DRAFT for teacher, counselor and legal review. Items marked [VERIFY] need checking against a primary source before publication.

AT A GLANCE

Big idea	Things will sometimes go wrong. What separates a small problem from a big one is a plan, calm roles, and a blame-free review afterward . The "hotwash" turns a bad day into a better system.
Grade bands	K-2 (the "Oops plan"), 3-5 (Detective and Fixer drill), 6-8 (full Bakery tabletop), 9-12 (full + roles, evidence, communication, root cause)
CSTA-style concepts	Networks and the Internet (cybersecurity: response and recovery), Impacts of Computing, Computing Systems (troubleshooting) (codes: see Concepts Lab Index, Part 4 [VERIFIED 2026-10-05])
Time	15-minute and 45-minute versions
Cost and tech	Printed inject cards, role badges, a timer, chart paper. No devices.
Teachers do it first	Yes. Part A about 25 minutes.

MAKE IT STICK

Try This Now (60 seconds, no prep): Say the four words with a hand motion each: DETECT, CONTAIN, RECOVER, LEARN. Then ask a neighbor: "A friend says their account is sending strange messages. What is the very first thing you tell them?" (Tell a trusted adult, stay calm, change the password from a clean device.)

01 The big idea

A **security incident** is an event that harms, or could harm, the safety of information or systems: a hijacked account, a lost device, a leaked file, a locked computer. **Incident response (IR)** is how people notice, limit and fix it, and then improve so it is less likely or less costly next time.

Real incident responders follow a simple loop. This demo uses four easy words as a nod to professional guidance such as NIST's Computer Security Incident Handling Guide (SP 800-61), whose Revision 2 (August 2012) phases are usually described as **Preparation; Detection and Analysis; Containment, Eradication and Recovery; and Post-Incident Activity**. NIST withdrew Revision 2 on April 3, 2025 and replaced it with Revision 3 (April 2025), which aligns incident response to the Cybersecurity Framework 2.0 functions [VERIFIED 2026-10-05: [csrc.nist.gov SP 800-61 Rev. 2](https://csrc.nist.gov/SP/800-61/Rev.2) and [Rev. 3](https://csrc.nist.gov/SP/800-61/Rev.3) pages]. The four plain words below are a classroom simplification, not NIST's wording:

FIELD NOTE	No. 1
DETECT (notice and understand) → CONTAIN (stop it spreading) → RECOVER (fix and restore) → LEARN (the hotwash) ...and Prepare before anything happens. The loop never really ends.	

A **hotwash** is a quick, structured after-action review held right after an exercise or event ("hot" because it is immediate). The term comes from military and emergency-response practice [VERIFY origin]. It asks: what was supposed to happen, what actually happened, why, and what will we do differently? Its heart is **blamelessness**: look for how the *system* made the mistake easy, not who to punish.

02 What a teacher should know (plain language)

2.1 WHY PRACTICE? (THE FIRE DRILL LOGIC)

NO ONE LEARNS TO evacuate during a fire. Drills build habits and expose gaps in a safe setting. A **tabletop exercise** is a discussion-based drill: people sit around a table (or desks), react to a scenario revealed in steps (called **injects**) and talk through decisions. No devices, no real systems, no stress about real consequences.

2.2 THE FOUR STAGES IN DETAIL

STAGE	KEY QUESTIONS	EXAMPLE MOVES
Prepare (before)	Who do we call? Where is the plan? Do we have backups and two locks?	Contact list, backups (Demo 10), updates (Demo 7), 2SV (Demo 6), practice drills
Detect	What happened? How do we know? Who noticed? How serious?	Collect reports, check logs or alerts, write a timeline, name the incident, decide severity
Contain	How do we stop it from getting worse?	Disconnect a device, change passwords from a clean device, sign out all sessions, pause the affected account
Recover	How do we fix it and get back to normal safely?	Remove the cause, restore from backup, re-enable systems, monitor for repeats
Learn	What went well? What didn't? What will we change?	Hotwash, action items with owners and dates

Also: **Communicate** throughout (who needs to know, what do we tell them) and **document** (a timeline: who, what, when).

2.3 KEY IDEAS FOR TEACHERS

- **Speed and calm beat perfection.** The first goal is to limit harm; panic and secrecy make incidents worse.
- **Fixing symptoms is not enough.** "We changed the password" is containment; "we found why it happened" is learning (Five Whys).
- **Evidence matters.** Do not delete messages and logs in a panic. Preserve a timeline and screenshots according to policy (never preserve illegal images; see the safety section).
- **Reporting is a duty and a help.** Many organizations must notify affected people or regulators after certain breaches; laws vary by place and sector [VERIFY legal duties; schools may have FERPA and state obligations; consult counsel].
- **Blamelessness is not "no accountability."** People make mistakes, so good systems make mistakes harder and recoverable. Honest reporting only happens when it is safe to report.
- **Roles reduce chaos:** one decides (Incident Commander), one investigates, one fixes, one communicates, one writes it down.

2.4 FIVE WHYS (A SIMPLE ROOT-CAUSE TOOL)

Problem: "The bakery's email sent junk to customers."

1. Why? Because someone else logged in as the owner.
2. Why? Because they had the owner's password.
3. Why? Because an employee typed it into a fake page.
4. Why? Because the message looked like it came from the email provider and had a "your account will be closed" deadline.

5. Why couldn't we stop it? Because there was no second lock and no habit of checking through a known route.

Fixes: turn on two-step verification, practice the Six P's (Demo 3), add a rule "always go to the provider's site yourself", set up an easy way to report weird messages.

2.5 METRICS (9-12)

Teams track **time to detect** and **time to contain** (sometimes abbreviated MTTD and MTTC or MTTR [VERIFY terminology]) as measures of improvement. The point of measuring is to learn, not to rank people.

2.6 COMMON MISCONCEPTIONS

MYTH	BETTER UNDERSTANDING
"Incident response is for big companies."	Families and classes have incidents, too. Everyone benefits from a plan.
"If we find the person who clicked, we've found the problem."	The click is rarely the root cause. Systems and habits are.
"Tell no one so we don't look bad."	Delay and secrecy make things worse and prevent help.
"Blameless means nobody is responsible."	Everybody is responsible for fixing the system.

03 Materials (zero to low cost)

- **Role badges** (index cards on string): Incident Commander, Detective, Fixer, Communicator, Scribe, plus optional Observer.
- **Inject Cards** (below), printed and folded.
- **Playbook cards** (one per team): the four stages with checklists.
- **Timeline sheet** (a big sheet of paper divided into time slots).
- **Hotwash worksheet** (end of file), chart paper, markers.
- A **Blame Jar** (a cup) and sticky dots or coins.
- A bell or timer.

04 The Drill: "Trouble at Maple Street Bakery" (fictional)

4.1 SETUP

The class is **Maple Street Bakery**, a small family bakery that takes orders by email and a website, with six staff. Everyone is a staff member except the teacher (the **Narrator**), who reads injects. Teams of 5 take roles. Each team has the same scenario at the same time; the teacher can run them in parallel.

4.2 PLAYBOOK CARD (EACH TEAM GETS ONE)

- **Detect:** What do we know? What do we not know? Who might be affected? How serious is it (low, medium, high)?
- **Contain:** What is the quickest safe step to stop it getting worse?
- **Recover:** How do we fix the cause and restore? How do we check it's really fixed?
- **Communicate:** Who needs to know? What do we say, and what don't we know yet?
- **Learn:** What worked? What didn't? What will we change?

4.3 THE INJECTS (READ ALOUD, ONE AT A TIME, WITH A TIMER)

Inject 1 (Detect). "9:00 a.m. A regular customer, Mr. Ortiz, calls: 'I got an email from the bakery offering free cupcakes if I click a link. It looks odd.'" *Teams decide: What is this? What do we check first? Who do we tell?*

(Teacher note: this is the first clue that the owner's email account may have been taken over.)

Inject 2 (Detect). "9:10 a.m. Two more customers report the same email. The owner says she did not send it. A staff member, Sam, says, 'Yesterday I clicked a link from the email provider saying my account would close, and I typed my password.'" *Teams decide: How serious? What do we say to Sam? (Use blameless language.) What is likely happening?*

Inject 3 (Contain). "9:20 a.m. The owner checks her email on her phone and sees 'new sign-in from a far-away place' alerts. The emails are still going out." *Teams decide: What is the fastest safe step? (Change the password from a clean device, sign out all sessions, turn on two-step verification. Do NOT delete the emails yet.)*

Inject 4 (Contain, complication). "9:35 a.m. The owner realizes the same email address is used for the bakery's bank account, the delivery app and the website's admin login." *Teams decide: What else could be affected? Who do we call? How do we protect those accounts? (Check other accounts, change passwords, contact the bank using the number on the card.)*

Inject 5 (Communicate). "10:00 a.m. A local newsletter writer calls: 'Is it true the bakery was hacked?' Customers are posting on social media." *Teams decide: What do we tell customers and the writer? (Short, honest, no blame: what happened, what we're doing, what they should do: do not click, delete the email, change their password if they clicked.) Who is the spokesperson?*

Inject 6 (Recover). "11:00 a.m. The password is changed and two-step verification is on. The owner notices the email account has a **forwarding rule** she did not create that sends copies of her mail to a stranger's address." *Teams decide: What is this? What should be removed? What else should be checked?*

(Remove unknown rules, review recovery email and phone, check connected apps, review recent activity, watch for repeat.) [VERIFY: forwarding rules are a commonly reported persistence technique; this is a defender's checklist item.]

Inject 7 (Learn). "End of day. The incident is contained. Time for the hotwash." Move to Section 5.

4.4 SCORING (COLLABORATIVE, LIGHT COMPETITION)

Teams earn **Calm Points** (1 each) for: using the roles, writing a timeline, containing before cleaning up, using blameless words, honest communication, and asking for help. When someone says "whose fault" or "how could they", a teammate drops a coin in the **Blame Jar** and the team **recasts** the sentence in systems language ("what let this happen?"), which earns a Calm Point. There is no penalty and no one is singled out: the jar is a language game, not a punishment. Teams compare timelines and decisions, not just scores.

05 The Hotwash

RUN THE HOTWASH IMMEDIATELY after the drill: about **13 minutes** in the 45-minute version, **3 minutes** in the 15-minute version.

Ground rules (read aloud): "We talk about **what happened**, not **who**. Everyone speaks. Facts first, opinions second. We end with actions."

The four hotwash questions (from after-action review practice):

1. **What was supposed to happen?** (The plan and expectations.)
2. **What actually happened?** (Use the timeline.)
3. **Why were they different?** (Use Five Whys, systems not people.)
4. **What do we sustain and what do we improve?** (Name 2 things to keep, 3 things to change.)

Action items: each improvement gets an **owner** (a role, not a name in this classroom version) and a **date**. Keep the list short; three good actions beat ten wishes.

Closing round: each student shares one word for how the drill felt, and one habit they will keep.

06 Part A: Teachers do it first (about 25 minutes)

RUN THE 15-MINUTE VERSION as participants in role groups (five roles, with extra as Observers). Then:

1. **Experience the speed (12 min).** Run the 15-minute version in role groups (extra people as Observers). Notice how quickly roles blur and how easily "who clicked?" enters.
2. **Hotwash (8 min).** Use the four questions and write down two actions.
3. **Reflect (5 min).** What is your own plan for a lost device, a hijacked account or an online bullying incident? Who is the contact, and is it written down?

07 Part B: Teaching it

15-MINUTE VERSION

MINUTES	WHAT HAPPENS
0-2	Hook: the fire drill analogy; show the four words (Detect, Contain, Recover, Learn).
2-4	Assign roles; read the playbook card.
4-11	Injects 1, 3 and 6 (about two minutes each); teams respond in writing on the timeline.
11-14	Mini hotwash: the four questions, with one "keep" and one "change".
14-15	Exit slip: "In an incident, the first thing I do is ____."

45-MINUTE VERSION

MINUTES	WHAT HAPPENS
0-5	Hook, vocabulary, roles.
5-10	Practice round: a tiny incident ("a classmate's device is lost") to learn the roles.
10-32	The full Bakery drill, Injects 1-6, 3 minutes each, narrator reads and teams decide.
32-45	The Hotwash: four questions, Five Whys, action items. Close with the Calm Points and one habit per student.

K-2 AND 3-5 ADAPTATION: THE "OOPS PLAN"

K-2: "STOP. TELL. FIX. Learn." Use the **Oops Plan hand signals**: Stop (Pause Palm), Tell (Tell Wave), Fix (rolling hands), Learn (point to head). Run three silly "oops" stories (a spilled juice on the library book; an app that made a sound; a game that asks for a password). Practice saying "Oops! I need to tell you something."

3-5: Detective (what happened?), Fixer (what do we do now?) and Teller (who do we tell?). Run Injects 1, 3 and 6 simplified. Hotwash question: "What went well? What do we do differently next time?"

08 Teacher script

Hook (2 minutes). "What do we do in a fire drill? Walk out, line up, count heads. Why practice? Because nobody thinks clearly in a real fire. Security problems are the same. Today we practice for an 'oops' in a bakery and then we do something real teams do after every drill: a hotwash. It's a quick meeting where nobody gets blamed and everybody gets better."

During injects. "Detective, what do we know and not know? Fixer, what is the quickest safe step? Communicator, what do we say? Scribe, write the time. Commander, you decide. If you're stuck: read the playbook card."

Blameless language coach. "I heard 'Sam messed up'. Let's turn it into 'The fake page looked very real. What would have helped Sam catch it?' Systems, not people."

Hotwash opener. "We're not here to judge. Nobody did badly. We're here to learn. What was supposed to happen? What did happen?"

Close. "The best teams aren't the ones that never have incidents. They're the ones that notice fast, act calmly, tell the truth and learn. If something ever goes wrong online, for you or a friend, it's not your fault. Telling quickly is the bravest and most useful thing you can do."

09 Kid-friendly analogies by band

BAND	ANALOGY AND ACTIVITY
K-2	The Oops Plan: Stop, Tell, Fix, Learn, like a fire drill. Hand signals. "Oops is a safe word."
3-5	Detective, Fixer and Teller. Mini drill with three injects. "What went well? What next time?"
6-8	The Bakery drill with roles, timeline and Calm Points. Hotwash in four questions.
9-12	Tabletop exercise and post-incident review. Severity and impact, evidence handling, legal and ethical communication, Five Whys, metrics, continuous improvement. Compare with a public incident report and its lessons learned [VERIFY a suitable public after-action report].

10 Discussion questions

1. What was the most useful first move, and why? Why is it wrong to focus on "who clicked"?
2. How did two-step verification and backups (Demos 6 and 10) change the story?
3. What should you say to customers: too little, too much, or just right?
4. (9-12) How do you balance speed of containment with preserving evidence?

11 Level-up challenge

- **Write your class's Incident Plan:** one page for "a lost device, a hijacked account, a mean message and a scary message", with who to tell and what to say.
- **Design a new inject** (for example, "a delivery app locks the bakery out") with the expected good responses, and run it with another team.
- **Compare with real guidance (9-12).** Read the executive summary of a public incident-handling guide (such as NIST SP 800-61 or CISA's playbooks) and map the phases onto the class drill [VERIFY].

12 Safety and ethics notes (defense only)

- **Fiction only.** Use the fictional bakery and its injects. Do not drill on real school systems, real incidents, or real students' accounts. Do not simulate attacks. Do not role-play as "the attacker".
- **No hunting for culprits.** If a real incident arises, follow the school's process; do not turn the drill into an investigation of a real student.
- **No real data.** Do not ask students to share real passwords, accounts, devices or incidents. If a student brings up a real incident, acknowledge it privately and follow your reporting process (see the Index box).
- **Evidence and illegal content.** For incidents involving sexual images, threats or exploitation, do **not** view, copy or forward the content; follow your school's disclosure procedure and legal reporting steps (Teacher Guide, "If a student discloses"). A tabletop drill is never a place to handle such material.
- **Psychological safety.** Some students have had real family or school incidents. Keep the tone calm; allow opting out of roles; avoid dramatic or frightening wording.

13 Assessment idea

- **Exit ticket.** "Name the four words of incident response and one blameless sentence." Rubric: 3 = Detect, Contain, Recover, Learn and a correct blameless sentence; 2 = names stages but blames; 1 = "call the police".
- **Timeline task.** Give the scrambled injects. Students reorder, mark where Contain occurred, and mark the root cause.

14 Sources and [VERIFY] list

- NIST SP 800-61 Revision 2 (2012) and Revision 3 (April 2025) "Incident Response Recommendations and Considerations for Cybersecurity Risk Management: A CSF 2.0 Community Profile"; Revision 2 (August 2012) was withdrawn April 3, 2025 [VERIFIED 2026-10-05: csrc.nist.gov].
- NIST Cybersecurity Framework 2.0 (2024) functions: Govern, Identify, Protect, Detect, Respond, Recover [VERIFIED 2026-10-05: NIST CSWP 29, dated February 26, 2024].
- CISA Incident and Vulnerability Response Playbooks and tabletop exercise packages (CISA Tabletop Exercise Packages, "CTEPs") [VERIFY].
- After-action review practice from the US Army and the Homeland Security Exercise and Evaluation Program (HSEEP) hotwash terminology [VERIFY].
- Blameless post-incident reviews (Google SRE book, Sidney Dekker's "Just Culture") [VERIFY].
- UK NCSC "Exercise in a Box" tabletop resources [VERIFY].

Student Handout: The Hotwash

Name: _____ Team: _____

The four words

DETECT (what happened?) → **CONTAIN** (stop it getting worse) → **RECOVER** (fix it) → **LEARN** (hotwash)

My role today

COMMANDER / DETECTIVE / Fixer / Communicator / Scribe (circle)

Timeline (Scribe fills in)

TIME	WHAT HAPPENED OR WHAT WE DECIDED
9:00	_____
9:10	_____
9:20	_____
10:00	_____

The Hotwash: four questions

1. What was **supposed** to happen? _____
2. What **actually** happened? _____
3. **Why** were they different? (Think systems, not people.)

4. What will we **sustain** (keep)? _____

What will we **improve**? _____

Blameless words

INSTEAD OF "SAM MESSED up," I can say: "The fake page looked real. What would have helped?" My own blameless sentence: _____

My action

One thing I will do this week: _____

Remember

- Notice fast. Tell the truth. Act calmly. Learn.
- If something goes wrong online for you or a friend, **it is not your fault.**
Pause. Check. Tell a trusted adult.

End of student handout.

TEACHER COPY ONLY:

Answers for Demo 13 (do not distribute to students)

FIELD NOTE

No. 2

WARNING: Teacher page. Remove before copying. The student handout ends on the page before this one. Print the handout pages only, or detach this page first. Do not post it, project it or send it home.

Where the bakery drill's answers show up on the handout:

- **Contain:** change the password from a clean device, sign out all sessions, turn on two-step verification (and do not delete the emails yet).
- **Learn:** the hotwash and the fixes.
- **Root cause (Five Whys):** there was no second lock and no habit of checking the source through a known route.
- **Blameless sentence examples:** "The fake page looked very real. What would have helped Sam catch it?"; "What made this easy to miss, and what can we change?"

Rubric reminders: exit ticket 3 = Detect, Contain, Recover, Learn plus a correct blameless sentence; 2 = names stages but blames; 1 = "call the police".